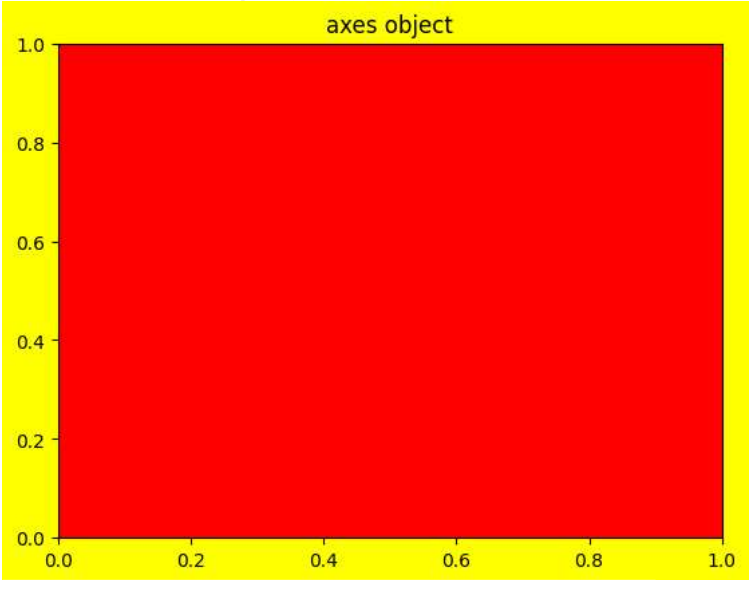
```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
ax.set_title("axes object")
```

```
Text(0.5, 1.0, 'axes object')
```



Create a coordinate system: left boarder, bottom left boarder, width,height. The values must be between 0 and 1

fig.add_axes([l,b,w,h])

is another way of creating an Axes object inside a Figure.

```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
ax1=fig.add_axes([0.4,0.4,0.3,0.3]) # added an axis object with left,bottom,height and width specified
plt.savefig("2D plot.pdf") # save graphics in pdf format
x=[1,2,3,4,5]
y=[1,4,9,16,25]
ax1.plot(x,y)
ax1.set(xlabel='x',ylabel='y',title='y=x^2')
fig.show()
```

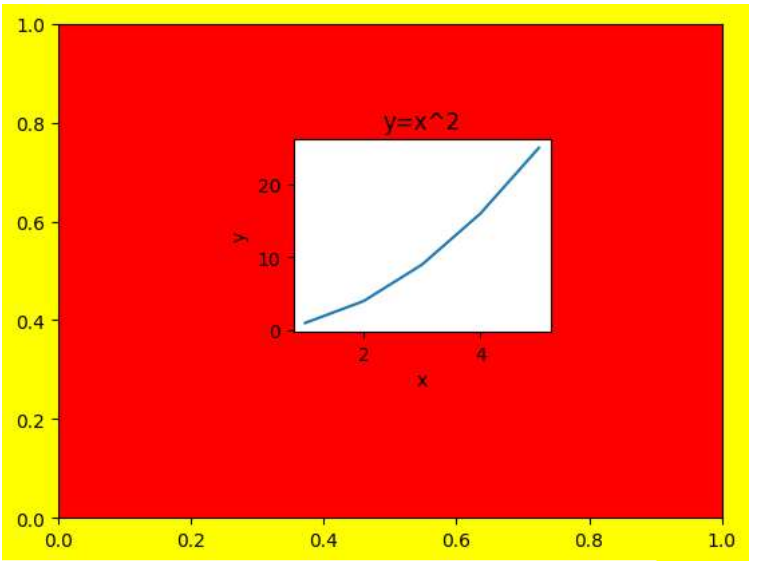
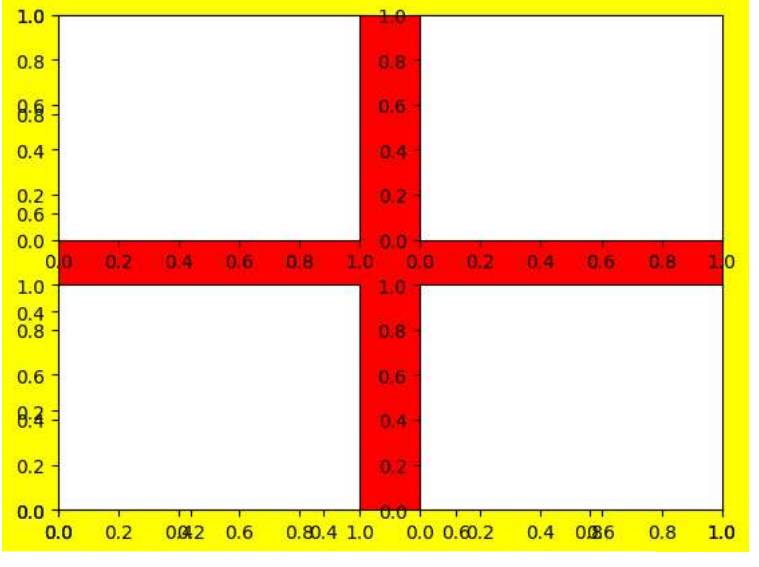
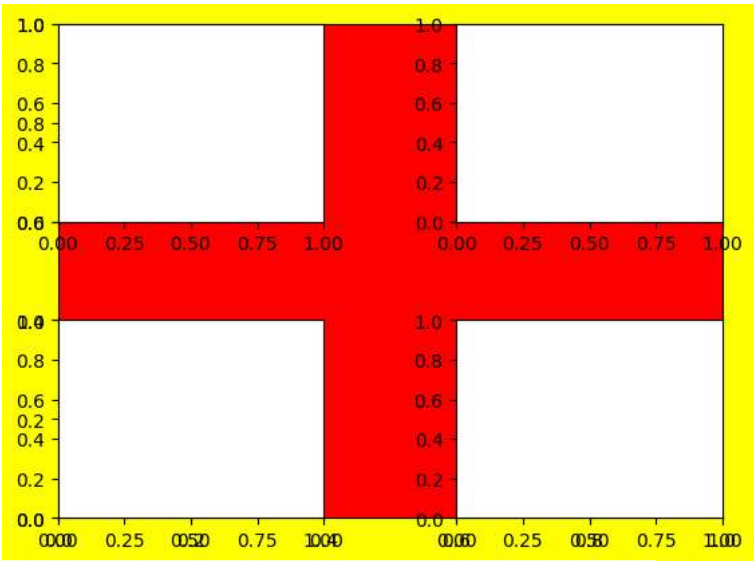


Fig.subplots(r,c) sets the position. creates subplots with r and c columns

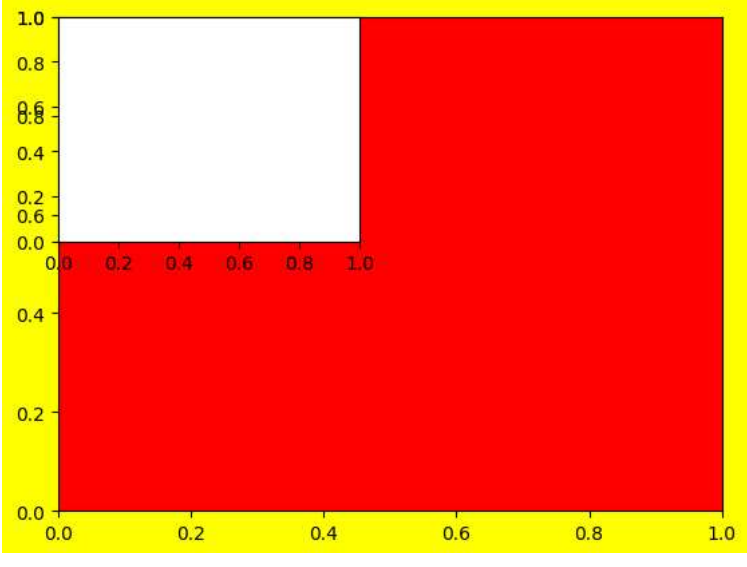
```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
#ax1=fig.add_axes([0.4,0.4,0.3,0.3])
ax=fig.subplots(2,2)
#ax=fig.subplots_adjust(wspace=0.5,hspace=0.5) # horizontal space and vertical space between subplots
```



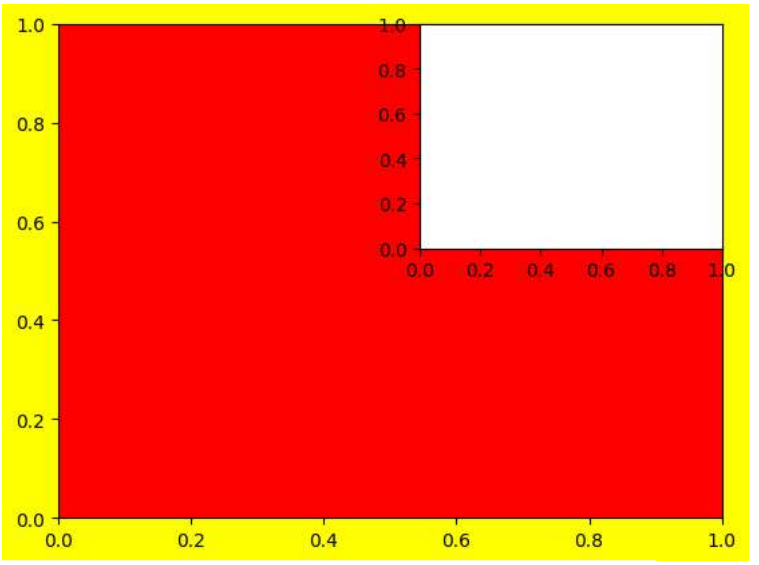
```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
#ax1=fig.add_axes([0.4,0.4,0.3,0.3])
ax=fig.subplots(2,2)
ax=fig.subplots_adjust(wspace=0.5,hspace=0.5) # horizontal space and vertical space between subplots
```



```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
#ax1=fig.add_axes([0.4,0.4,0.3,0.3])
#ax=fig.subplots(2,2)
ax=fig.add_subplot(2,2,1)
#ax=fig.add_subplot(2,2,2)
#ax=fig.add_subplot(2,2,3)
#ax=fig.add_subplot(2,2,4)
```



```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
#ax1=fig.add_axes([0.4,0.4,0.3,0.3])
#ax=fig.subplots(2,2)
#ax=fig.add_subplot(2,2,1)
ax=fig.add_subplot(2,2,2)
#ax=fig.add_subplot(2,2,3)
#ax=fig.add_subplot(2,2,4)
```



Write a Python program using Matplotlib that:

1. Creates a Figure containing four subplots in a 2 × 2 arrangement.
2. Plots:
   - (y=x^2)
   - (y=sqrt(x))
   - (y=sin(x))
   - (y=cos(x))
3. Gives suitable titles and axis labels.
4. Uses tight_layout() to improve spacing.
5. Saves the figure as a PDF file.
6. Displays the figure.

```
import numpy as np
import matplotlib.pyplot as plt

# Generate x values
x = np.linspace(0, 10, 200)

# Create Figure and 2 × 2 subplots
fig, ax = plt.subplots(2, 2)

# Subplot 1: y = x^2
ax[0, 0].plot(x, x**2)
ax[0, 0].set_title("y = x²")
ax[0, 0].set_xlabel("x")
ax[0, 0].set_ylabel("y")

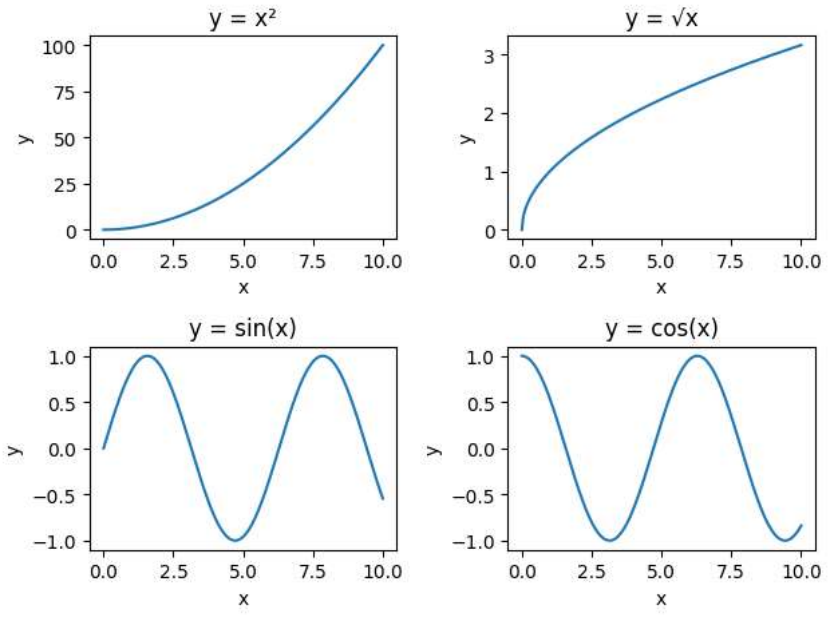
# Subplot 2: y = sqrt(x)
ax[0, 1].plot(x, np.sqrt(x))
ax[0, 1].set_title("y = √x")
ax[0, 1].set_xlabel("x")
ax[0, 1].set_ylabel("y")
# Subplot 3: y = sin(x)
ax[1, 0].plot(x, np.sin(x))
ax[1, 0].set_title("y = sin(x)")
ax[1, 0].set_xlabel("x")
ax[1, 0].set_ylabel("y")

# Subplot 4: y = cos(x)
ax[1, 1].plot(x, np.cos(x))
ax[1, 1].set_title("y = cos(x)")
ax[1, 1].set_xlabel("x")
ax[1, 1].set_ylabel("y")
```

```
# Adjust spacing between subplots
fig.tight_layout()

# Save the figure
fig.savefig("four_functions.pdf")

# Display the figure
plt.show()
```




```
import matplotlib.pyplot as plt
import numpy as np

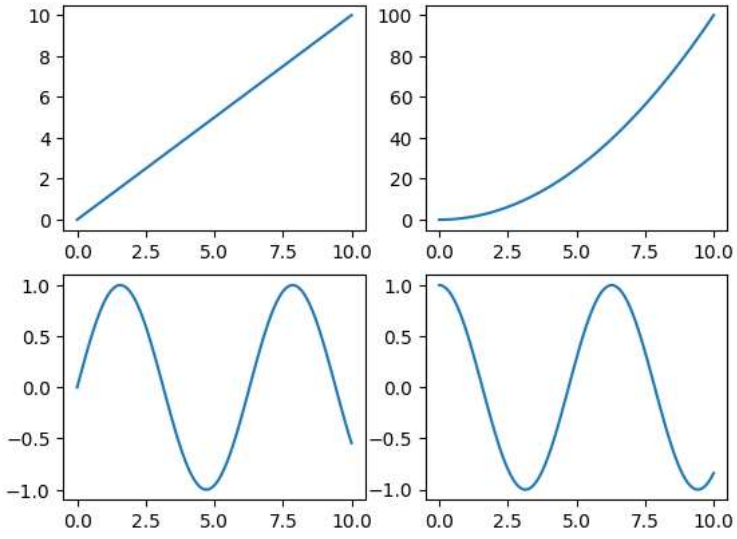
x = np.linspace(0, 10, 100)

fig = plt.figure()

ax1 = fig.add_subplot(2, 2, 1)
ax2 = fig.add_subplot(2, 2, 2)
ax3 = fig.add_subplot(2, 2, 3)
ax4 = fig.add_subplot(2, 2, 4)

ax1.plot(x, x)
ax2.plot(x, x**2)
ax3.plot(x, np.sin(x))
ax4.plot(x, np.cos(x))

plt.show()
```

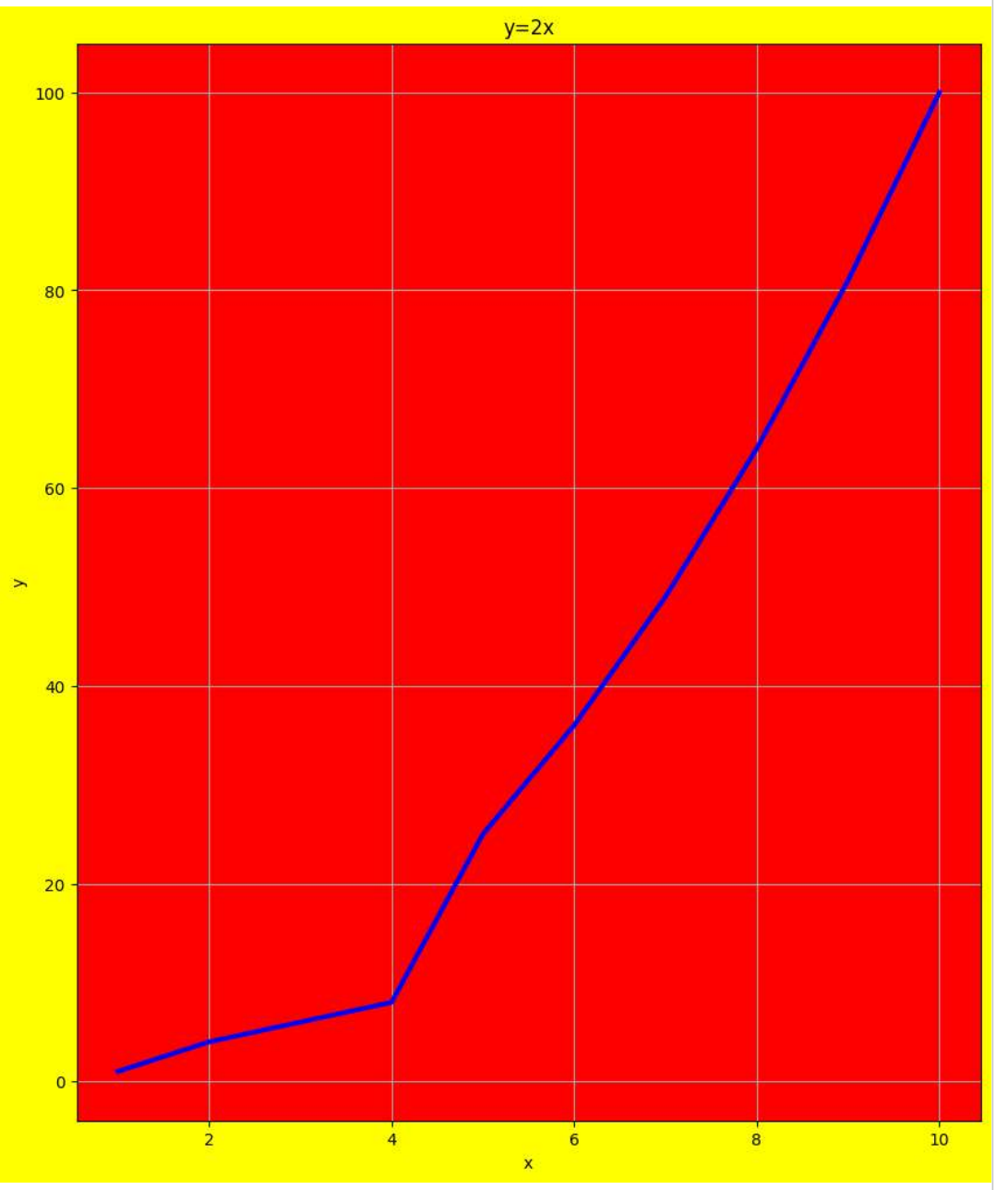


```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
ax.plot(x,y,lw=3,color='b')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('y=2x')
ax.set_xlim(0,20)
ax.set_xticks([0,2,4,6,8,10,12,14,16,18,20])
ax.set_ylim(0,200)
ax.text(4,120,'point')
ax.grid()
plt.show()
```

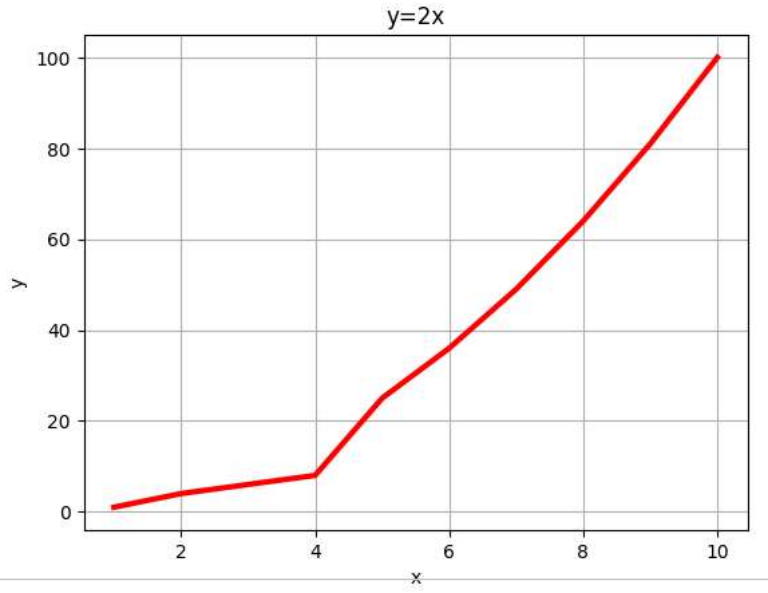



```
import matplotlib.pyplot as plt
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
fig.set_figheight(12)
fig.set_figwidth(10)
ax.plot(x,y,lw=3,color='b')
```

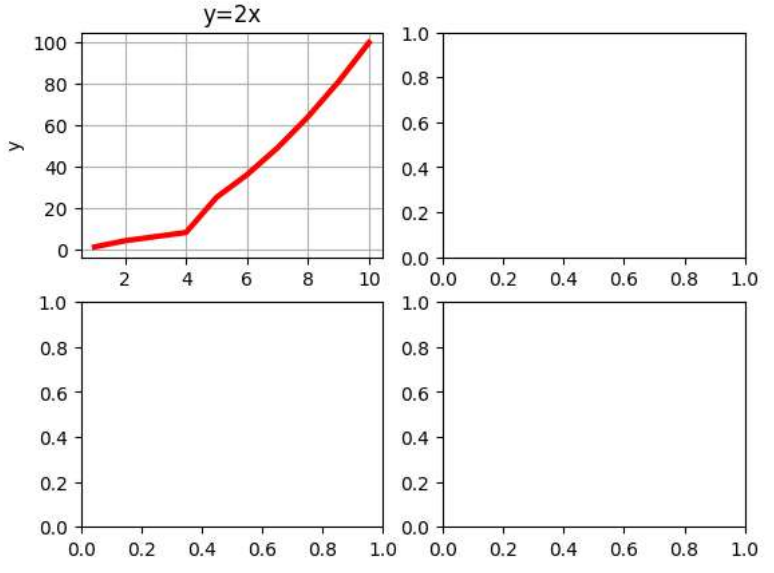
```
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('y=2x')
ax.grid()
plt.show()
```




```
import matplotlib.pyplot as plt
x=[1,2,3,4,5,6,7,8,9,10]
y=[1,4,6,8,25,36,49,64,81,100]
fig=plt.figure()
ax=fig.subplots()
ax.plot(x,y,lw=3,color='r')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('y=2x')
ax.grid()
plt.show()
```




```
import matplotlib.pyplot as plt
x=[1,2,3,4,5,6,7,8,9,10]
y=[1,4,6,8,25,36,49,64,81,100]
fig=plt.figure()
ax=fig.subplots(2,2)
ax[0][0].plot(x,y,lw=3,color='r')
ax[0][0].set_xlabel('x')
ax[0][0].set_ylabel('y')
ax[0][0].set_title('y=2x')
ax[0][0].grid()
plt.show()
```




```
fig=plt.figure()
ax=fig.subplots()
fig.set_facecolor('yellow')
ax.set_facecolor('red')
fig.add_subplot(2,2,3)
```

```
<Axes: >
```

